is the total number of existing cases with respect to the entire population.

Depending on the spread of infectious disease in the community, they may be classified into different types.

- Endemic diseases are those which are constantly present in a particular area. Typhoid fever is endemic in most parts of India.
- Epidemic disease is one that spreads rapidly, involving many persons in an area at the same time. Example: Epidemic of Dengue in 2017.
- A pandemic is an epidemic that spreads through many areas of the world involving very large numbers of persons within a short period. Example: H1N1 Influenza outbreak in 2009. Ebola outbreak in 2014-2016 in West Africa was the largest in history and first ever epidemic, affecting multiple countries.
- If a particular disease occurs only occasionally, it is called a sporadic disease. The most commonly occurring sporadic diseases in India are Diphtheria and Hepatitis A and E.

Severity or duration of a disease

Another useful way of defining the scope of a disease is in terms of its severity or duration.

- An **acute** disease is one that develops rapidly but lasts for a short time.
- A **chronic** disease develops more slowly, and the body's reactions may be less severe, but the disease is likely to be continual or recurrent for long periods.
- A disease that is intermediate between acute and chronic is described as a subacute disease.

Facts about Fever:

Fever is as more healthful than harmful. An experiment with vertebrates shows that fever increases the rate of antibody synthesis. Increased temperatures stimulate the activities of T cells and increase the effectiveness of interferon. Fever appears to enhance phagocytosis. Fever almost never occurs as a single response; it is usually accompanied by chills. The explanation lies in the natural physiological interaction between the thermostat in the hypothalamus and the temperature of the blood. For example: If the thermostat has been set (by pyrogen) at 102°F but the blood temperature is 99°F, the muscles are stimulated to contract involuntary (shivering) as a means of producing heat. In addition, the vessels in the skin constrict, creating a sensation of cold and the piloerector muscles in the skin develops 'goose bumps'.



- A latent disease is one in which the causative agent remains inactive for a time but then becomes active to produce symptoms of the disease.

12.1.4 Interaction between Microbes and Host

Pathogen is a microorganism which causes disease.

Pathogenecity is the ability of a pathogen to produce disease.

Virulence is the degree of pathogenecity of a microorganism. Virulence is not generally attributable to a single property

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